

Why Scores Are NOT the Final Answer

The goal of judging at ISEF is to identify the best projects. In reaching this goal two very different activities are required: (1) interviews with the finalists, and (2) discussions with your fellow judges. Neither of these by itself is sufficient.

Professional Scientific/Engineering Conferences

Think about what you do when you attend your own professional scientific or engineering conferences. At any conference with 1700 presenters there will be a relatively small number of plenary and invited talks compared with poster sessions. While the same considerations apply to both, let's talk about the poster sessions because the analogy here is so obvious. Conference attendees come to poster sessions in order to converse with (that is, to interview) the scientists presenting their work. Why? Because you want to find out what is new and important, both in general and for your own work.

There are too many posters to see them all, so you choose a limited set.

Those interviews are an important source of information. Even though you read the abstracts of many posters in advance of the conference, the interviews are how you learn first-hand about the work by the presenter. You can ask questions directly to learn what they have done, and to ensure that you understand what the presenter has tried to communicate.

You make assessments of each poster, informally, perhaps even without realizing it. Some were disappointing duds, many were good presentations of solid work, and a precious few will be remembered long after the conference is over.

Reflecting Later on the Posters

After the poster session is over you get together with other scientists, perhaps in the hallway or perhaps over a beer, and talk about what you've seen. You may have thought that you saw the best poster in the room even though you only interviewed a small fraction of them. Then another attendee tells you of a different presentation that you didn't see. Discussing this unseen presentation, as well as the relative merits of projects seen by both of you, causes you to reevaluate your earlier impressions.

The most important work at conferences is often not in the formal presentation sessions, but rather in mutually beneficial discussions afterwards. Which presentations do you talk about? Certainly not all of them. No, you talk about the ones that are of most interest to you, the ones you feel are, in some sense, "best." The more attendees you speak with, the more you learn about worthwhile presentations you didn't see – and those that you did.

At real scientific conferences, most talks and posters are rather ordinary, but some will pique your interest and make an impression, while a very few may be so important that they will impact your own research program and that of other scientists.

All of these experiences correspond precisely to what you will find at ISEF.

The ISEF “Poster Session”

Similarly, at ISEF your interviews are at formal “poster sessions.” The posters constructed by the students will emulate, and not infrequently surpass, the quality of many posters presented at professional scientific conferences. It is during these “poster sessions” when you will form your own opinion about each project that you see.

Just as at your own scientific meetings, there are too many projects at ISEF, even within your own category, for you to see all of them, so you are assigned a limited set. The tool ISEF uses here is the set of ISEF subcategories within each category. Each finalist characterizes their project’s specific focus, while judges identify their primary area of expertise. The interview assignments then make use of that information to match students and judges, so as to optimize the experiences of both.

Finally, and analogously to your assessment of professional presentations, you will formalize your assessment of each project you interview by assigning a numeric score. Importantly, however, this numeric score is only the first step in determining awards.

Reflecting Later on the Posters

At the end of the afternoon, you and your fellow judges will get together and talk about what you have seen. Thanks to the comments by other judges you will change your opinion of some projects you have seen which arise during the caucus, not to mention all that you have not.

Which projects do you talk about? Certainly not all of them. In the caucus room you will be presented with a caucus support tool that allows everyone in the room to focus attention on the projects which, by virtue of assigned numeric scores, have already been determined to be, in some sense, “best.”

But the numbers are not the final word. They are merely the starting point for your discussions. As the judges collectively discuss each project’s merits, a general sense of the room will develop as to the merits of each, which sometimes will be quite different from what the numeric scores suggested. The caucus process at ISEF provides a simple means by which candidate projects for each award can be rapidly identified, discussed, and evaluated.

Conclusion

The important message here is that the computer spitting numbers out at you only begins the process of picking award winners. Intelligent people discussing the relative merits of the best projects will always make better choices than dumb computers looking at numbers generated by judges working in isolation.

Points of Emphasis

1. You will formalize your evaluation of each project by scoring each interview.
2. All scores will be processed into a form presented in the caucus enabling all judges to focus their discussion on the projects that are likely most award-worthy.
3. The processed scores do not determine awards. They merely inform the discussion and subsequent voting on award candidates.